Math Minds: A Stepwise Solution Generator for Engineering Mathematics Problems Using OCR

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Abstract. Engineering Mathematics (EM1–EM4) are core subjects in the Mumbai University curriculum, yet many students struggle with its complex problem-solving methods. This project acts as a platform for studying mathematics, resolve doubts and making learning easier. Students can either type the required problem or upload questions via screenshots or photos, which is then processed using Optical Character Recognition (OCR) to extract mathematical expressions. The extracted equations are converted into LaTeX format and processed using a symbolic mathematics engine to generate step-by-step solutions. Unlike traditional problem-solving tools that provide only final answers, the proposed system focuses on generating detailed intermediate steps to help students understand the solution process. The system then generates step-by-step solutions using rule-based algorithms and symbolic mathematics engine, helping students grasp concepts rather than just answers. The platform also offers topic-specific practice for all the modules. The project intends to function as a customized digital instructor by fusing AI with an interactive interface, lowering academic challenges and enhancing EM1–EM4 students' problem-solving abilities

Keywords: Optical Character Recognition (OCR),Stepwise solutions ,Digital tutor, LaTeX.

1 Introduction

Rapid scientific and engineering advancements in the modern world necessitate accurate and efficient solutions to challenging mathematical problems. Numerous technical disciplines, including computer science, electronics, mechanical design, and system engineering, are built on mathematics. However, lengthy analytical derivations, symbolic manipulations, and complex mathematical problems involving Fourier series, Laplace transforms, differential equations, and matrix computations can be challenging for both professionals and students. Manual fixes for these issues take a lot of time and could result in human error.

By offering an intelligent platform that can comprehend and solve mathematical expressions step-by-step, the suggested system "MathMinds: A Stepwise Solution Generator for Engineering Mathematics Problems using OCR" seeks to close the gap between conventional problem solving and computational automation [2][5]. Both

text-based and image-based inputs are supported by the system. Optical Character Recognition (OCR) methods are used to process image inputs, extracting mathematical expressions and transforming them into structured formats appropriate for symbolic computation.

The implementation creates an end-to-end processing pipeline by integrating multiple Python-based technologies. Mathematical expressions can be identified from both printed and handwritten images using PaddleOCR and EasyOCR [12]. Pix2Text allows for the structured representation of mathematical expressions by converting the extracted equations into LaTeX format. The SymPy library carries out symbolic operations like algebraic simplification, differentiation, integration, and transform evaluation after the equation has been transformed into a machine-readable format [6]. While MathJax is used to render the final solutions in an understandable mathematical format, supporting libraries like Pillow and NumPy help with image preprocessing and numerical operations. Through the integration of computer vision, artificial intelligence, and symbolic mathematics, the platform serves as both an automated solver and a learning tool that provides step-by-step solutions. Instead of depending only on definitive solutions, this method fosters conceptual understanding.

All things considered, the suggested platform shows how contemporary computational methods can make solving challenging mathematical problems easier while preserving interpretability and transparency. The system offers a scalable framework for intelligent mathematics learning and automated problem solving in engineering and research settings by fusing OCR technologies with symbolic computation.

2 Related Work

The main motivation behind this project emerged after research on the automated mathematical problem-solving systems and learning platforms. We studied previous projects such as SOLVE IT(MATH SOLVER) and Wolfram alpha these works showcased us how automation reduces manual problem-solving efforts while also simultaneously teaching its users. These systems, showed strong symbolic computation capabilities and relying in solving complex mathematical problems, but they either lacked support for image-based inputs or have low accuracy for image-based inputs[2].This limited their usability for students who relied on their handwritten notes, question papers or textbook images. These highlighted a gap to us in the existing systems, which motivated us to create a more robust solutions which would solve these problems.

Advancements in educational technology emphasized the role of Intelligent Tutoring Systems (ITS) in improving learning outcomes. Systematic reviews and meta-analysis reported that the ITS platforms enhance students' performance through adaptive instructions and self-paced learning environments [5][4]. Research on web-based mathematics instruction further reinforced these findings, demonstrating that interactive and technology-driven learning environments enhance problem-solving skills, engagement, and accessibility across diverse learner groups [1]. These studies collectively emphasized that effective educational systems must move beyond static content delivery toward learner-centric and adaptive designs.

Previous research faced challenges for accurate recognition of mathematical problems. Traditional OCR techniques fail to preserve the two-dimensional structure of mathematical notations. To address this issue, recent studies proposed expression trees and convolutional sequence networks which are a type of structured representations of mathematical problems, and this helps maintain the mathematical validity and operators' hierarchy during equation generation [12][11]. These helped improve reliability and interpretability of machine generated equations.

Symbolic computation tools enabled the students to connect theoretical derivations with computational verification, which further enhanced the analytical and reduced conceptual gaps [7][3]. These findings reinforced importance of step-by-step derivations rather than black-box numerical outputs.

Recent studies on multimodal mathematical reasoning highlighted the potential of combining visual, textual and symbolic inputs for solving various mathematical problems [10]. At the same time, the research on handwritten mathematical expressions showed progress while using encoder-decoder architecture, graph neural networks and self-supervised attention mechanisms, while acknowledging persistent challenges in generalization and interpretability [9][8].

3 Methodology

The proposed system follows a systematic approach to aid students in resolving Engineering Mathematics problems (EM1–EM4) while promoting conceptual understanding. At first, the user chooses the type of math module and the problem can be given as text or as an image by taking a screenshot.

To make the input images easier to read, they are preprocessed. This makes it possible to use OCR techniques to accurately extract mathematical expressions[4]. The extracted content is then changed into the right LaTeX format so that it can be structured and processed further.

Then, the system uses standard university-level methods to match the problem to a set of pre-defined solution strategies. Solutions are produced sequentially, akin to classroom derivations, guaranteeing clarity and accuracy [6].

Finally, the results are formatted into readable mathematical expressions and displayed through a web interface [12]. The platform also organizes problems into topic-based modules to support guided practice and self-paced learning, helping students better understand complex mathematical concepts.

3.1 Proposed System

The proposed system follows a structured architecture to ensure modularity, clarity, and efficient processing of Engineering Mathematics problems. The system handles both textual and image based inputs and then provide structured step-by-step solutions.

User Interface Layer. The user interface layer acts as the interaction module between the user and the system. It allows the users to either manually enter mathematical

expressions or they can upload images containing handwritten or printed equations as shown in Fig 1.

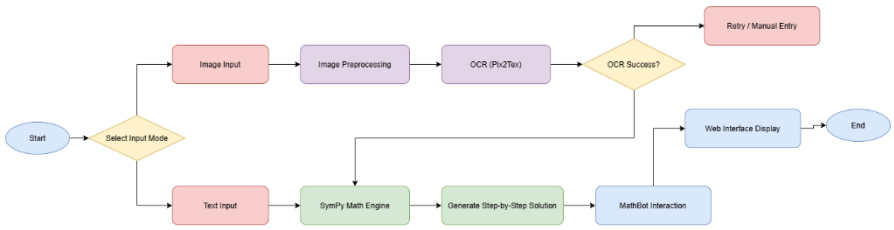

Fig. 1. Proposed System.

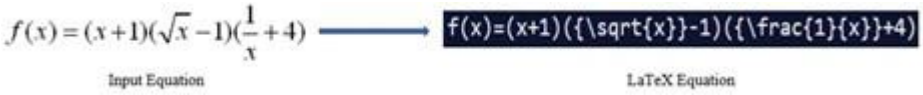

Fig. 2. LaTeX Conversion of Mathematical Equation

The system supports two modes of user input: text-based and image-based. When a user enters a mathematical expression in text form, it is first converted into a standardized LaTeX representation to ensure uniformity and accurate symbolic interpretation which is shown in Fig.2. In case of image-based input, the image is first processed to extract the mathematical expression, which is later converted to its corresponding LaTeX form[11]. The system converts all equations to its LaTeX form to establish a consistent intermediate representation for further processing.

Backend Layer. It is the central control unit of the proposed system. It receives input from either the user interface or the OCR module through a request handler, which is responsible for managing request routing, validation and error handling. This ensures that only well-formed and interpretable inputs proceed to subsequent stages to be processed.

An equation parser analyzes the extracted text to determine the specific mathematical operation involved, such as integration, Laplace transform, or matrix computation. Based on the analysis, the extracted expressions are arranged into a standardized format, which is suitable for further computation.

Math Engine Layer. It performs the core computational tasks. Using symbolic computation techniques, the engine executes the required mathematical operations. Additionally, a step-by-step logic generator decomposes the solution into intermediate

stages, enabling users to understand the complete problem-solving methodology rather than just the final result [7].

Finally, the Output Module formats the computed result and its corresponding step-by-step explanation into a readable mathematical form. The formatted output is sent back to the user interface, where it is displayed using mathematical rendering tools to ensure clarity, correctness, and ease of interpretation from Fig.3.

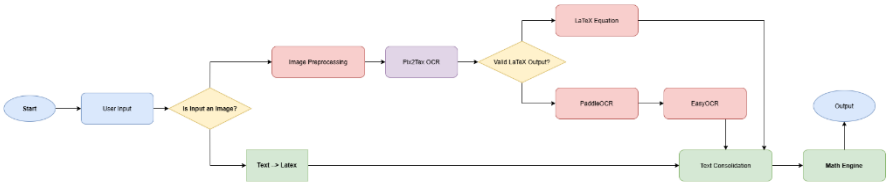

Fig. 3. Working of OCR Models

OCR Models.

PaddleOCR. It is an OCR framework for general text detection and recognition that is based on deep learning. It works well with structured mathematical expressions and legible printed text that are frequently found in scanned documents, PDFs, and textbooks. Text regions are first identified and then transformed into machine-readable strings using PaddleOCR detection–recognition pipeline. It might, however, have trouble with intricate spatial structures found in sophisticated mathematical notation.

Pix2Tex. It is a specialized model created especially to convert mathematical expression images into LaTeX format. To capture the spatial arrangement of symbols, it employs an encoder-decoder architecture with a vision transformer-based encoder and a sequence generation decoder. Because of this, it is better at identifying complicated equations that contain fractions, matrices, superscripts, and subscripts.

EasyOCR. It is used for more difficult inputs, which include examples of handwritten equations or images taken under low-light conditions. It helps improve the overall robustness of the system by being able to interpret text that is not well-formed (unusual) or very noisy from an image taken with a smart device. In addition, while its ability to comprehend complex mathematical structures is limited, it provides a fallback for cases when other OCR methods are not able to yield results.

Chatbot. The chatbot operates through a layered workflow comprising the frontend, backend task logic, and the Gemini API, as illustrated in the Fig.4. The user first submits a query or selects the predefined steps through the chatbot interface. The input is captured and forwarded to the backend for processing. In the backend task layer, the system extracts the relevant question, and then constructs a query by combining the

user's query and the system information. This prompt is then sent to the Gemini API for further processing.

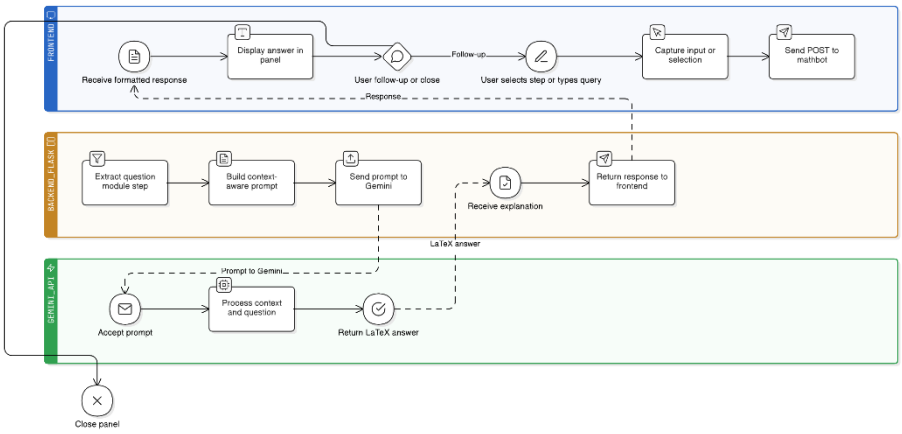

Fig. 4. Workflow Of Gemini API.

Upon receiving the generated prompt the Gemini API analyses it and then generates a LaTeX formatted explanation. This generated response is then forwarded to the backend, which validates and formats it properly before forwarding it to the frontend. The system also supports follow-up interactions, so the user can enquire further about the steps. This workflow continues iteratively until the user themselves choose to close the panel this ensures that there is a seamless interaction between the user and the system.

3.2 System Overview

This system is organized around the Mumbai University (MU) curriculum for the subjects Engineering Mathematics I, II, III, IV. Each of these subjects are having six different modules being taught in the undergraduate engineering program. These modules each cover a vast range of concepts spanning from differential equations, matrices, probability, Laplace transform etc. This helps the users to navigate concepts systematically. Also, in each module there is a dedicated formula sheet which is provided which provides the users a quick access to important formulas and key concepts for their revision and reference. This feature helps the users to quickly recall the fundamental concepts while solving problems, which further improves their problem solving capabilities.

3.3 Handling OCR Errors

The correctness of the mathematical expression extracted during the OCR stage determines the accuracy of symbolic computation in the suggested system. In real-world situations, handwriting variations, image noise, or intricate mathematical layouts can occasionally cause OCR models to misinterpret symbols. The system incorporates a user validation step prior to the computation stage in order to address this problem.

The interpreted equation appears on the user interface once the OCR module has extracted the equation and transformed it into the appropriate LaTeX representation. If any recognition errors are found at this point, the user has the option to manually edit to fix the equation. The equation is sent to the symbolic computation engine for step-by-step solution generation only upon user confirmation. This human-in-the-loop verification method guarantees greater reliability of the produced mathematical derivations and greatly lessens the impact of OCR recognition errors on the final solution. The system maintains accuracy and flexibility for various input formats, including handwritten notes, scanned question papers, and screenshots, by permitting user correction.

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4 Dataset Used

We have utilized the IM2LATEX-100K dataset, which is a large-scale, prebuilt benchmark dataset designed for image-to-LaTeX generation tasks. It contains approximately 100,000 paired samples, where each sample consists of a rendered mathematical formula image and its corresponding LaTeX source code. Unlike the traditional datasets, IM2LATEX-100k does not contain predefined numerical features, instead the model learns these features automatically by receiving raw inputs:

Image Features: Each sample contains rendered formula images. Pixel level data is extracted using Convolutional Neural Networks (CNNs) while model training, which helps the model capture the structural information of the Mathematical expressions such as the spatial alignment, symbols and Mathematical layout.

Textual Features: The output is recorded as a sequence of LaTeX tokens, these tokens are then converted into their corresponding embeddings, where each token represents a learned feature from the high dimensional space.

Feature Dimensionality: These are dependent on the model used and vary based on the architecture used in the CNN encoder and sequence decoder. Hence, the dataset itself does not have a fixed feature count.

The key dataset Characteristics and Statistics used in this study is summarized in Table 1.

Table 1. Dataset Characteristics and Statistics.

Attributes	Details
Total Samples	103,558 image-formula pairs
Training Samples	90,000
Validation Samples	5000
Test Samples	5000
Image Resolution	160x320 pixels
Average Formula Length	30-40 LaTeX Tokens

5 Model Evolution

Table 2. Performance Comparison of OCR Models.

Metrics	PaddleOCR	EasyOCR	Pix2Tex
Character Recognition Accuracy (%)	92.4	78.1	96.8
Precision (%)	78.6	72.3	95.9
Recall (%)	91.2	76.4	95.1
F1-Score (%)	90.1	78.8	95.5
LaTeX Parsing Success Rate (%)	81.3	74.5	92.7

As shown in Table 2 , Pix2Tex performs better than the other general OCR models in every expression-level statistic, making it ideal for Engineering Mathematics tasks involving complex notations.

Although PaddleOCR has high character-level accuracy, it has trouble maintaining mathematical structure, especially when dealing with multi-line or nested formulas.

EasyOCR works well for simple text recognition, but it struggles to parse complex mathematical symbols and portray them accurately. Higher LaTeX parsing success rates directly correlate with improved symbolic computation reliability and step-by-step solution generation.

5.1 Experimental Setup

The suggested system uses both pretrained models and experimentally evaluated baseline models for OCR and image-to-LaTeX. In order to assess the efficacy of deep learning architectures for mathematical expression recognition, samples from the IM2LATEX-100K dataset were used to implement and train CNN+RNN and CNN+LSTM. These experiments yielded the performance metrics reported for these baseline models.

However, the obtained accuracy was lower compared to specialized mathematical OCR systems. Therefore, pretrained models such as PaddleOCR, EasyOCR, and Pix2Tex were integrated into the final system pipeline. The benchmark results recorded in the corresponding research publications and official implementations serve as the foundation for the performance metrics presented for these models. In order to give a comparative overview of the efficacy of various OCR techniques for tasks involving the recognition of mathematical expressions, these metrics are included.

6 Results

The main goal of the proposed system is to help students learn Engineering Mathematics by correctly interpreting the equations they send in and giving them step-by-step symbolic solutions. It also needs to keep the math structure, work with different types of input, and work with symbolic computation. Fractions, matrices, integrals, and summations are common two-dimensional layouts in engineering math problems. Pix2Tex changes pictures of equations into structured mathematical forms while keeping the spatial hierarchy and operator precedence. This makes it better than regular OCR engines, which treat equations as linear text and often lose structural information.

Table 3. Table captions should be placed above the tables.

Model	Accuracy	Precision	Recall	F-1 score
TesseractOCR	67	65	58	61
InftyReader	87	88	82	85
Im2LaTeX	92	91	89	90
Mathpix	94	94	93	93
CNN + RNN	81	70	77	78
CNN + LSTM	80	78	79	78
PaddleOCR	88	88	84	86
EasyOCR	79	80	76	78
Pix2Tex	92	93	91	92

As shown in Table 3 there are other high-accuracy systems like MathPix, but they need cloud access or proprietary APIs, which makes it hard to reproduce and use them for academic work offline. PaddleOCR and EasyOCR, on the other hand, are good opensource options that work well on printed and semi-structured equations.

They also make the system faster and more scalable. Students send in equations in a number of different ways, such as typed text, screenshots, and handwritten notes. No single OCR model works best in all situations, so combining general-purpose OCR engines with a math-aware recognition model makes it more robust against changes in handwriting, image quality, and formatting.

The symbolic computation engine (SymPy) used in the system works with the outputs of these models to make it possible to find solutions step by step. This method keeps things clear and helps the platform reach its educational goal of being a learning tool instead of a black-box solver.

6.1 Experimental Results

After opening the MathMinds interface, as student can navigate through semester-wise engineering Mathematics module and can select their required subject.

After selecting a subject, the student is presented with all the major engineering Mathematics topics. From here on the student can choose a specific module such as

Eigen vector and values, complex integration, Z-Transform, Inverse Z-transform, Linear programming etc.

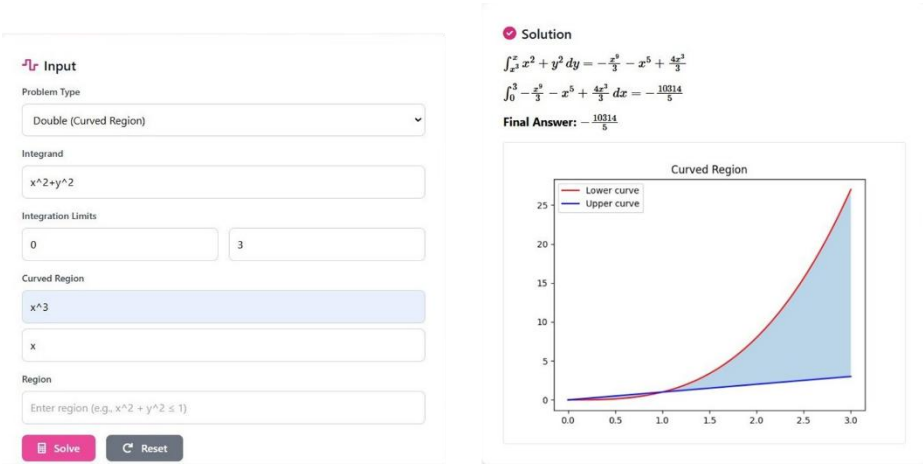

Fig. 5. Input and final solution for the double integration problem.

The Fig.5. shows the step-by-step solution generated by the system for evaluating a double integral over a curved region.

$$\int_0^3 \int_{x^3}^{x} (x^2 + y^2)\, dy\, dx \tag{1}$$

The equation gets converted into its corresponding LaTeX format. The LaTeX format of the equation is then forwarded to the backend logic generator. In the first step the backend system performs integration with respect to 'y' which is the inner integral.

$$\int_{x^3}^{x} (x^2 + y^2)\, dy = \left[x^2 y + \frac{y^3}{3}\right]_{x^3}^{x} \tag{2}$$

After applying the limits y = x^3 to y = x:

$$= \left(x^3 + \frac{x^3}{3}\right) - \left(x^5 + \frac{x^9}{3}\right) = \frac{4x^3}{3} - x^5 - \frac{x^9}{3} \tag{3}$$

In the second step the system performs integration with respect to 'x' which is the outer integral.

$$\int_0^3 \left(\frac{4x^3}{3} - x^5 - \frac{x^9}{3}\right) dx = \left[\frac{x^4}{3} - \frac{x^6}{6} - \frac{x^{10}}{30}\right]_0^3 \tag{4}$$

Apply limits to the equation as x = 3 to x = 0:

$$27 - \frac{729}{6} - \frac{59049}{30} = -\frac{10314}{5} \tag{5}$$

The final result of the problem is then showcased onto the interface.

The final output presents the evaluation of a double integral over a curved region defined by the functions y = x3 and y = x within the interval $0 \leq x \leq 3$. The shaded plot illustrates the region of integration, and the corresponding analytical solution is obtained by sequential integration with respect to y and x. The resulting negative value reflects the reversal of upper and lower bounds over the specified domain.

7 Conclusion

Using OCR and symbolic computation, the suggested system offers an intelligent learning platform for resolving Engineering Mathematics (EM1–EM4) problems. Instead of depending solely on final answers, the platform allows students to obtain structured step-by-step solutions that improve conceptual understanding by supporting both text-based and image-based inputs. Problem interpretation and solution generation are enhanced when OCR models are integrated with symbolic computation. All things considered, the system shows how AI-driven tools can enhance mathematics education by making difficult problem solving simpler while preserving interpretability and transparency.

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